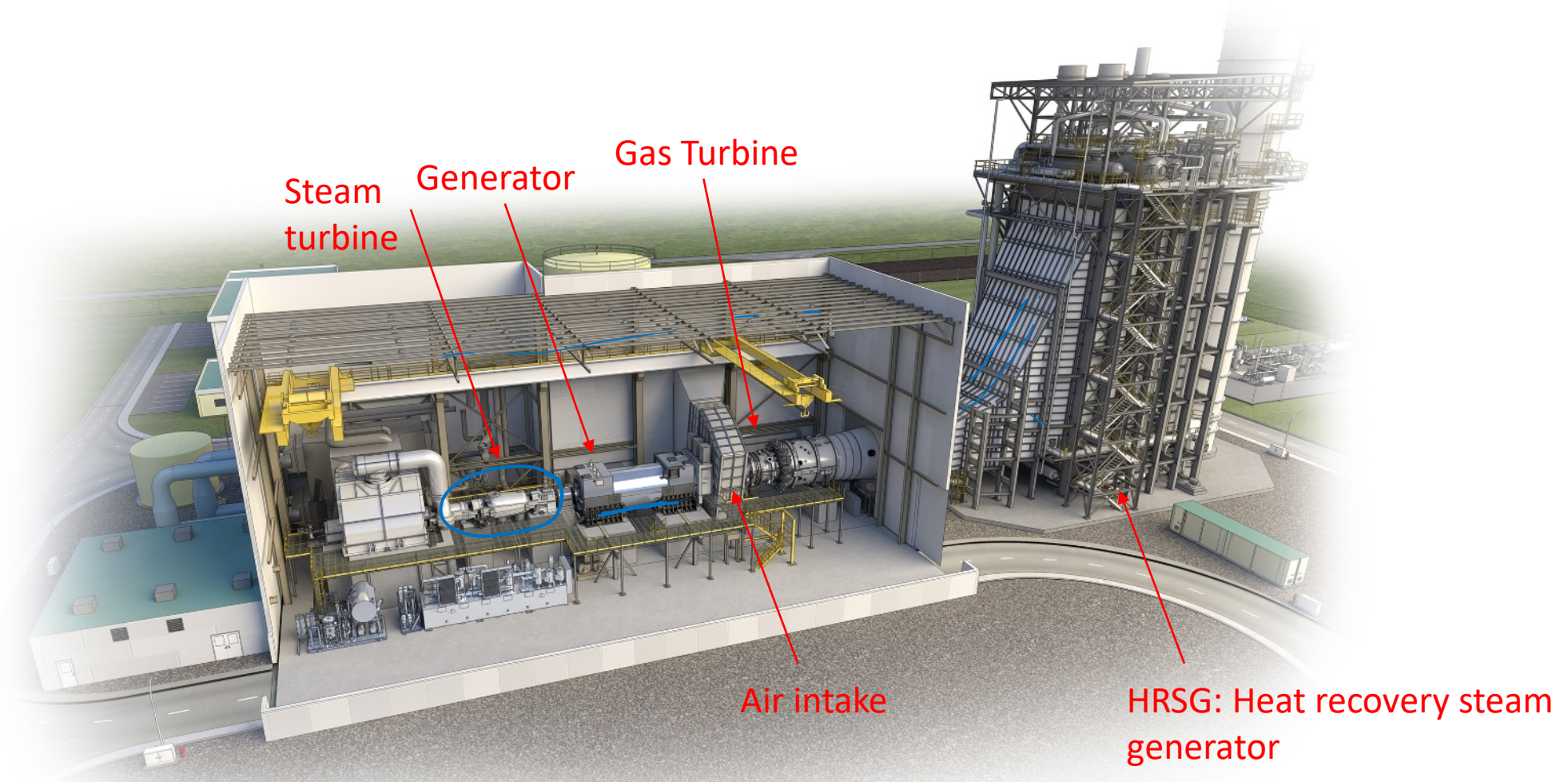




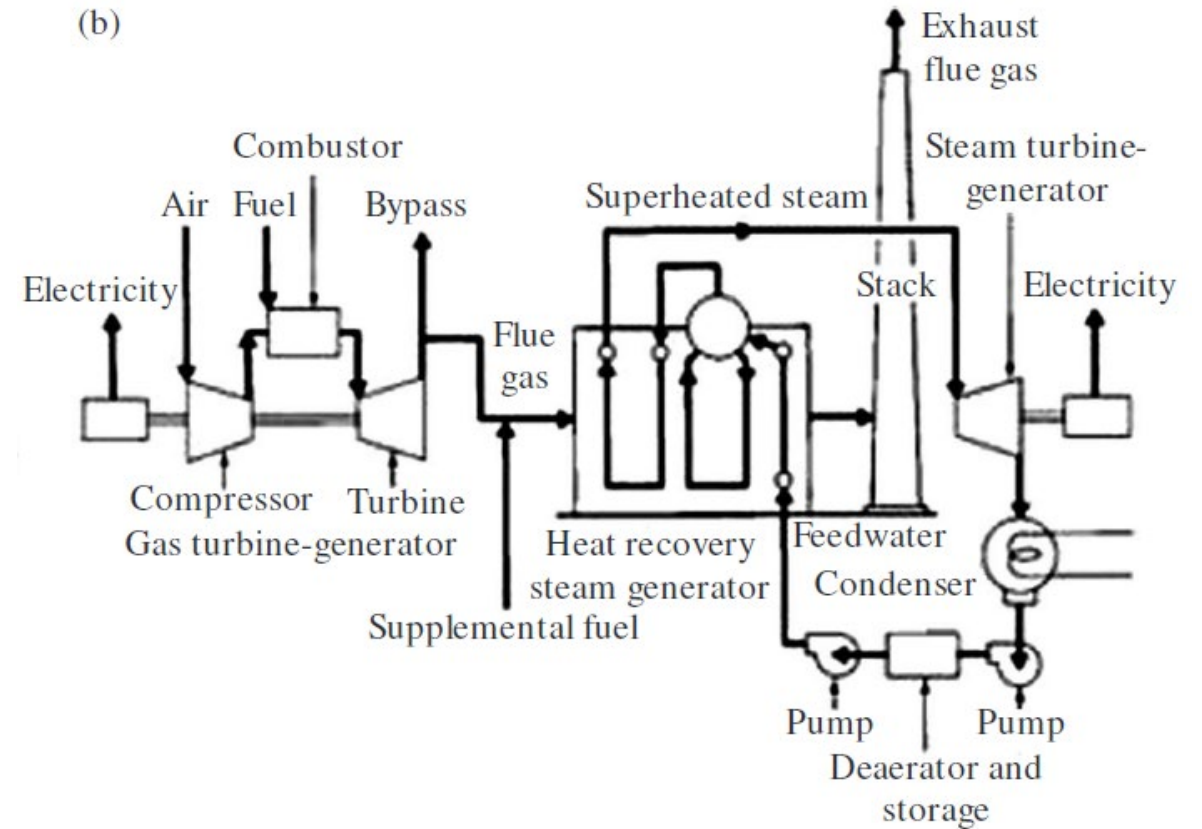
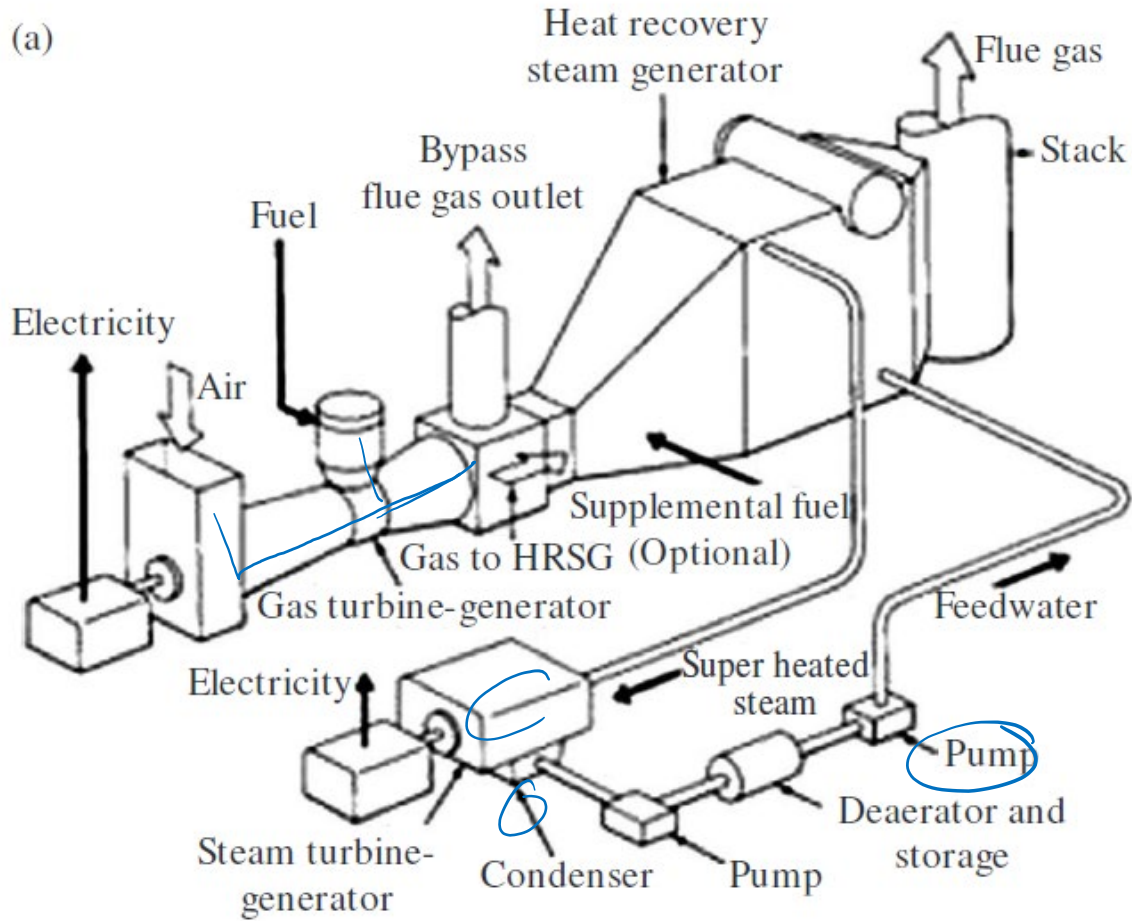
ME 300: Engineering Thermodynamics

Example: Combined Cycle Analysis

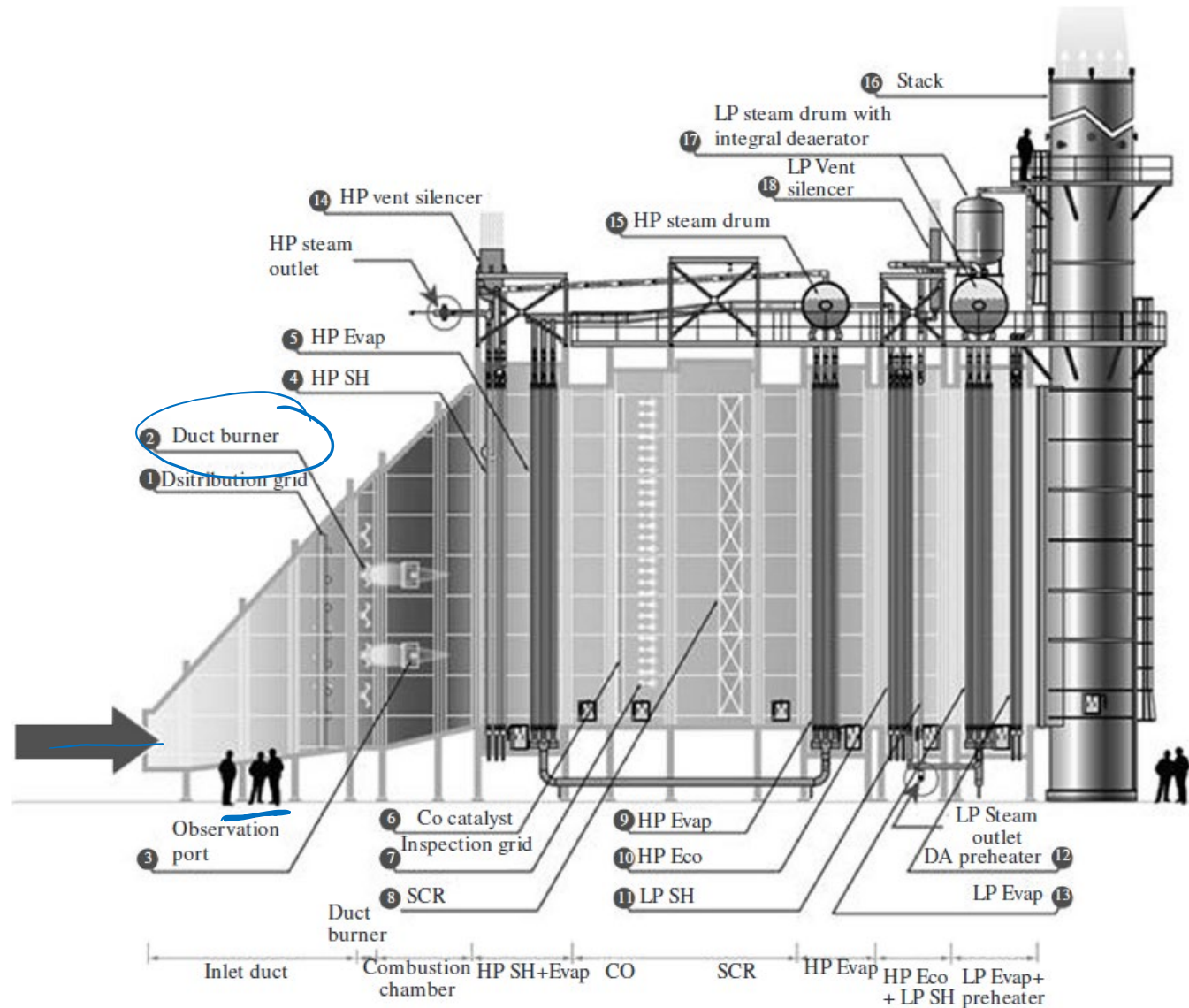
Combined cycle power generation



Combined cycle power generation



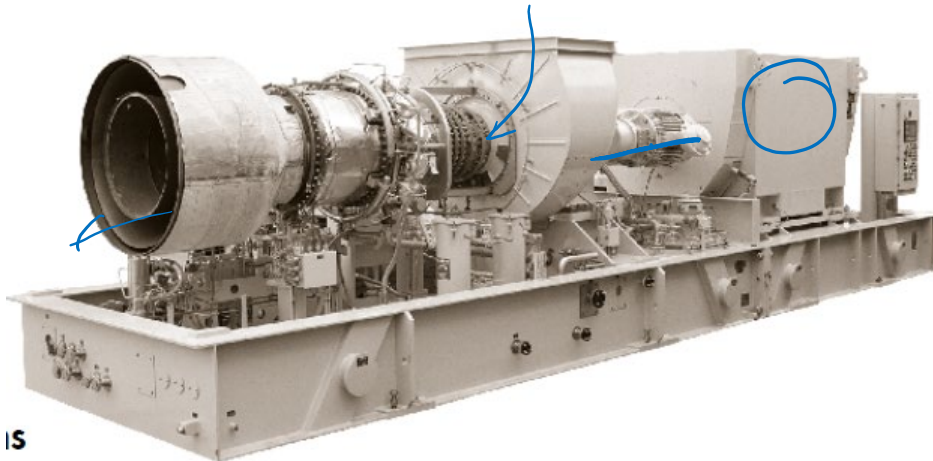
Combined cycle power generation



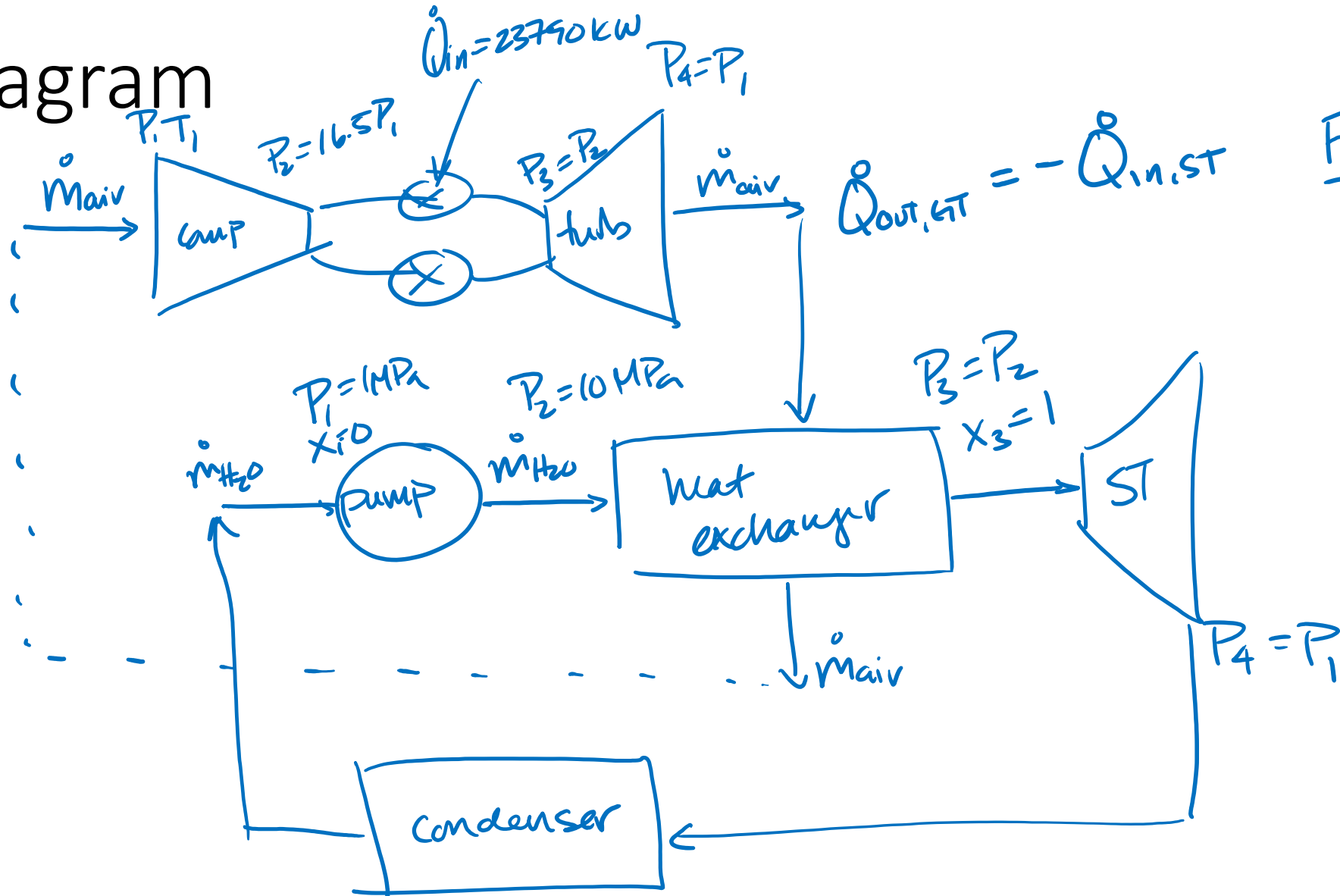
Example

The Taurus 70 gas turbine has a pressure ratio of 16.5, a mass flow of 26.4 kg/s, and a heat addition rate of 23790 kW. Assume an ideal air standard cycle with $T_1=300$ K and $P_1=0.1$ MPa ($R=287$ J/kg-K, $\gamma=1.4$, $c_p=1001$ J/kg-K). The exhaust from this engine powers an ideal Rankine cycle that runs between 1 MPa and 10 MPa with saturated liquid before the pump and saturated vapor coming out of the boiler. Using this information, calculate:

- The water mass flow rate of the Rankine cycle
- The net work of the combined cycle
- The efficiency of the combined cycle



Diagram



Find

: $\dot{m}_{H_2O}$

: $\dot{W}_{netcc} = \dot{W}_{GT} + \dot{W}_{ST}$

: η_{cc}



Will the cycle efficiency be higher or lower with combined cycles?

A. Higher

B. Lower

$$\eta_{cc} = \frac{\dot{W}_{net}}{\dot{Q}_{in}} = \frac{\dot{W}_{GT} + \dot{W}_{ST}}{\dot{Q}_{in}}$$

GT ↗

Solving strategy

$$\begin{aligned}\rightarrow \dot{W}_{net,cc} &= \dot{W}_{GT} + \dot{W}_{ST} \\ &= (\dot{W}_2 + 3\dot{W}_4)_{GT} + (\dot{W}_2 + 3\dot{W}_4)_{ST}\end{aligned}$$

$$\rightarrow \eta_{cc} = \frac{\dot{W}_{net,cc}}{\dot{Q}_{in}}$$

$$\begin{aligned}\rightarrow \dot{m}_{ST} \rightarrow \dot{Q}_{in,ST} &= -\dot{Q}_{out,GT} \\ &= \dot{m}_{ST} (h_3 - h_2) \\ \Rightarrow \dot{m}_{ST} &= \frac{\dot{Q}_{in,ST}}{h_3 - h_2}\end{aligned}$$

comp, pump, turb: $\Delta ke = \Delta p \tau = 0$
 $\dot{Q} = 0$

$$\dot{W} = -\dot{m} \Delta h$$

$$\rightarrow \text{air: } \dot{W} = -\dot{m}_a \Delta T$$

$$\rightarrow \text{steam: } \dot{W} = -\dot{m} \underbrace{\Delta h}_{\text{tables}}$$

Assumptions

Brayton

- air standard (ideal gas)
- ideal cycle (rev.)
- $C_p, \gamma = \text{const}$
- Steady flow

Rankine

- ideal cycle (rev.)
- S.C.S.
- Steady flow

Brayton cycle states

State 1 : $T_1 = 300 \text{ K}$ // $P_1 = 0.1 \text{ MPa}$

State 2 : $P_2 = 16.5 P_1 = 1.65 \text{ MPa}$

: $T_2 = T_1 (P_2/P_1)^{\gamma-1/\gamma} = 300 (16.5)^{0.4/1.4} = 668.3 \text{ K}$ //

State 3 : $\dot{Q}_3 = 23790 \times 10^3 \text{ W} = \dot{m}_{\text{air}} c_p (T_3 - T_2)$

$T_3 = T_2 + \frac{\dot{Q}_3}{\dot{m} c_p} = 668.3 + \frac{(23790 \times 10^3)}{(26.4)(1001)} = 1568.5 \text{ K}$ //

State 4 : $P_4 = P_1 \rightarrow T_4 = T_3 (1/P_2)^{\gamma-1/\gamma} = 1568.5 (1/16.5)^{0.4/1.4}$
 $= 704.1 \text{ K}$ //

Work/heat/efficiency of Brayton cycle

$$\begin{aligned}\dot{W}_{GT} &= \dot{W}_2 + \dot{W}_4 = -\dot{m}c_p(T_2 - T_1) - \dot{m}c_p(T_4 - T_3) \\ &= -(26.4)(1001) [(668.3 - 300) + (704.1 - 1568.5)] \\ &= 13110.14 \text{ kW}\end{aligned}$$

$$\dot{Q}_{in} = 23790 \text{ kW}$$

$$\eta_{GT} = \frac{\dot{W}_{GT}}{\dot{Q}_{in}} = 0.551$$

$$\dot{Q}_{out} = \dot{Q}_{in} - \dot{W}_{net} = -\dot{Q}_{in,ST}$$

$$\dot{Q}_{in,ST} = 10679.86 \text{ kW}$$

Rankine cycle states

State 1 : $P_1 = 1 \text{ MPa}$
 $x_1 = 0$ } Table D.2 $h_1 = h_f = 762.52 \text{ kJ/kg}$
 $s_1 = s_f = 2.1381 \text{ kJ/kg-K}$

State 2 : $P_2 = 10 \text{ MPa}$
 $s_2 = s_1 = 2.1381 \frac{\text{kJ}}{\text{kg-K}}$ } Table D.2 $\rightarrow s_2 < s_f \rightarrow \text{liquid}$
Table D.4B $\rightarrow h_2 = 773.055 \text{ kJ/kg}$

State 3 : $P_3 = 10 \text{ MPa}$
 $x_3 = 1$ } Table D.2 $\rightarrow h_3 = h_g = 2725.5 \text{ kJ/kg}$
 $s_3 = s_g = 5.6160 \text{ kJ/kg-K}$

State 4 : $P_4 = 1 \text{ MPa}$
 $s_4 = s_3 = 5.6160 \frac{\text{kJ}}{\text{kg-K}}$ } Table D.2 $\rightarrow s_f < s_4 < s_g \rightarrow \text{mix}$
 $x_4 = \frac{s_4 - s_f}{s_g - s_f} = 0.78$ $h_4 = x_4 h_g + (1 - x_4) h_f = 2333.89 \text{ kJ/kg}$

Steam mass flow and work

$$\dot{m}_{ST} = \frac{\dot{Q}_{in,ST}}{h_3 - h_2} = \frac{10679.86 \text{ kW}}{(2725.5 - 773.055) \frac{\text{kJ}}{\text{kg}}} \rightarrow \dot{m}_{ST} = 5.47 \text{ kg/s}$$

$$\begin{aligned}\dot{W}_{ST} &= {}_1\dot{W}_2 + {}_3\dot{W}_4 \\ &= -\dot{m}_{ST} [(h_2 - h_1) + (h_4 - h_3)] = 2084.5 \text{ kW}\end{aligned}$$

Combined cycle work and efficiency

$$\dot{W}_{cc} = \dot{W}_{GT} + \dot{W}_{ST}$$

$$= 13110.14 + 2084.5 \Rightarrow 15194.62 \text{ kW}$$

$$\eta_{cc} = \frac{\dot{W}_{cc}}{\dot{Q}_{in}} = \frac{15194.62}{23790} \Rightarrow 63.87\%$$